

Quick-start on ICsprout Calibre LVS Flow



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Complete Setups in Calibre LVS Runset

- Before executing Calibre LVS flow, you need prepare the following files:
 - The CDL netlist of your design
 - The corresponding layout GDS of your design
 - The subcircuit.spice
 - The Calibre_LVS_Runset
 - Hcell file

Complete Setups in Calibre LVS Runset

- Here are the settings for metallization option 1P7M_6lc_1T4Mc as an example:

```

=====
/**OPTION 2: Define Total Metal. The value can be 10,9,8,7,6,5,4. Default value is 10.

#define TOTALMETAL 7
=====
/**OPTION 3: Define number of top metal.The value can be SINGLE or DOUBLE(Upper Case).
/**Default value is double.
/**If you use double top metal process, set this option to DOUBLE (Upper Case).
/**If you use single top metal process, set this option to SINGLE (Upper Case).
/**If you set TOPMETAL to 10, you must set option TOP_METAL_NUM to double, single top
/**metal for 1P10M is not allowed!
/**If you set TOPMETAL to 4, you must set option TOP_METAL_NUM to single, double top
/**metal for 1P4M is not allowed!

#define TOP_METAL_NUM SINGLE
=====
/**OPTION 4: Define WIDTH of TM2.The value can be 2,4 or 15. Default value is 4.
/**2: TM2 is defined as T2M2.
/**4: TM2 is defined as T4M2.
/**15:TM2 is defined as T8M2.

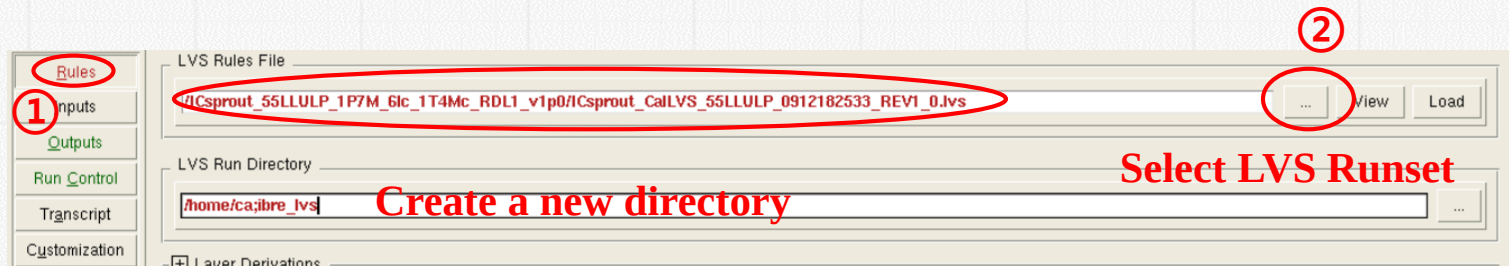
#define TM2_TYPE 4
=====
/**OPTION 5: Define WIDTH of TM1.The value can be 2,4 or 15. Default value is 4.
/**2: TM1 is defined as T2M1.
/**4: TM1 is defined as T4M1.
/**15:TM1 is defined as T8M1.
/**If TM1_TYPE is defined 4, then TM2_TYPE shold be 4 or 15, cannot be 2.
/**This option only effect in "TOP_METAL_NUM DOUBLE" condition.

#define TM1_TYPE 4
=====
/**OPTION 4: Define RC EXTRACT or not.The value can be TRUE or FALSE(Upper Case).
/**FALSE(Upper Case):it is only for LVS check.
/**TRUE (Upper Case):it is for LVS+RC flow.

#define RC_EXTRACT FALSE
=====
  
```

Run Calibre LVS by GUI

- Here are the settings for metallization option 1P7M_6lc_1T4Mc as an example:



Calibre LVS GUI Screenshot 1:

- 1** Inputs (selected in the left sidebar)
- 2** LVS Rules File: /ICsprout_55LLULP_1P7M_6lc_1T4Mc_RDL1_v1p0/ICsprout_CalLVS_55LLULP_0912182533_REV1_0.lvs
- 2** LVS Run Directory: /home/calibre_lvs (with red text "Create a new directory" and "Select LVS Runset")



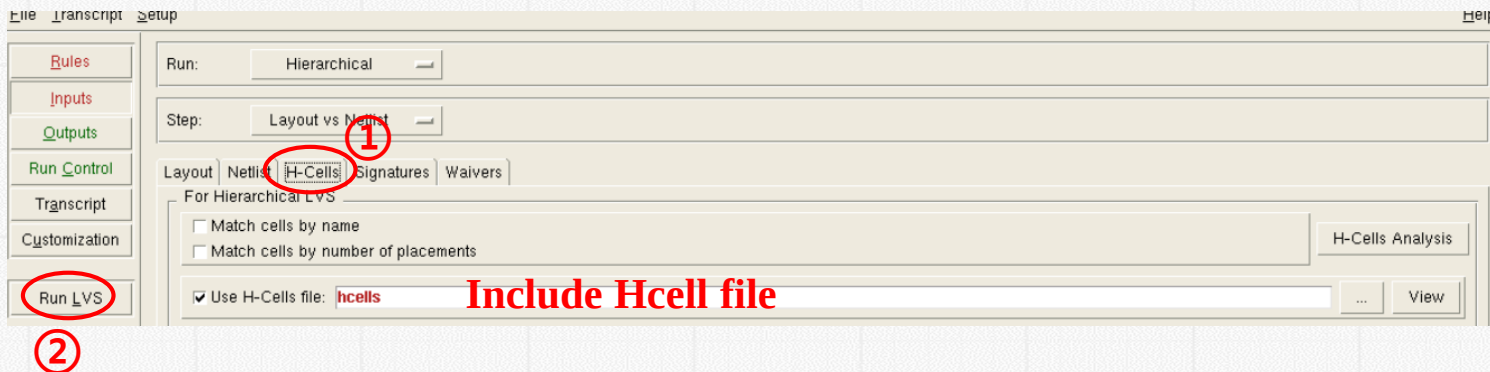
Calibre LVS GUI Screenshot 2:

- 1** Inputs (selected in the left sidebar)
- 2** Run: Hierarchical
- 3** Layout vs Netlist (selected in the top bar)
- 2** Layout (selected in the bottom bar)
- 3** Netlist (selected in the bottom bar)
- Format: SPICE
- Spice Files: (empty field)
- Export from schematic viewer: ☐

Select GDS/CDL/subcircuit.sp file

Run Calibre LVS by GUI

- Here are the settings for metallization option 1P7M_6lc_1T4Mc as an example:



Thanks